Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/11/2008
PI Name:	Lackner, James R. Ph.D.		
Project Title:	Somatosensory Suppression and Prevention of Post-Flight Re-Entry Disturbances of Posture and Locomotion		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02454-9110	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSBRI NNH04ZUU003N Human Health in Space
Start Date:	08/01/2005	**End Date:**	07/31/2008
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-NA00701		
Performance Goal No.:			
Performance Goal Text:			

Our overall goal was to develop procedures for alleviating post-flight disturbances of posture and locomotion experienced by astronauts. Our research has shown that either non-mechanically supportive contact of the hand with an environmentally fixed surface or pressure cues from a mobile, hand-carried gyroscope stabilizes postural control in a normal 1g environment, a rotating environment, and parabolic flight. Our strategy was to improve the effectiveness of the portable haptic device and to extend it to situations involving walking. Our original plan was to investigate three avenues of potential improvement: 1) miniaturizing the sensory gyroscope, 2) comparing the gyroscope to a Tactor vest, and 3) providing subjects training with both devices.

At the beginning of this project, we were directed to accelerate aspects of our work that would lead to the development/definition of countermeasures and would enhance collaborative team interactions with other NSBRI balance prosthesis projects. As a result, we spent the first year of the project focusing on research to evaluate the tactile

Task Description:	sensory encoding and control of passive body sway. We examined how the latency and pattern of postural stabilization are affected by touch contact of the hand with an environmentally fixed surface during quiet stance and whether position or velocity cues about body sway are keys to establishing stabilization. Our results showed that natural finger contact forces encode body sway velocity, and that the sensed finger forces in turn control the position of the center of foot pressure with a very short latency. These results specified requirements for the motion sensing and stimulus delivery components of a mobile, synthetic system for tactile alleviation of post-flight disturbances of posture and locomotion. Our Year 2 and 3 work evaluated whether body sway and locomotion could be stabilized by a tactor vest device or a few tactors attached to the hand or other body part delivering synthetic tactile cues that provide naturally timed position and/or velocity cues. We also evaluated the time course of postural stabilization and destabilization and whether immobilizing the arm is destabilizing. The results of these studies provide the basis for designing apparatus and procedures for suppressing re-entry disturbances after spaceflight and hastening sensory- motor readaptation to 1g, in the context of ISS, lunar and Mars missions. This research is most relevant to Risk #14, “Impaired Sensory-Motor Capability to Perform Operational Tasks After Landing and Throughout Re-Adaptation” and Risk #13 “Impaired Sensory-Motor Capability to Perform Operational Tasks During Flight, Entry, and Landing” in the NASA Bioastronautics Roadmap. This project advanced the research from the Countermeasure Readiness Level of 4 to a level of 5-6 and a Technology Readiness Level of 3-4.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The haptic cueing theory and technology which we develop in this project can potentially serve as mobility aids to people with neuromotor gait/balance disorders and as a spatial disorientation aid for pilots. We currently have AFOSR funding to develop pilot aids, and we have applied for NIH funding for clinical applications.
Task Progress:	Year 3 1) Tests were performed to determine whether fingertip touch cues would destabilize posture when the biceps muscle of the contacting arm was mechanically vibrated. Such vibration elicits reflexive contraction of the biceps. (We had earlier shown that light touch of the hand suppresses completely the otherwise destabilizing effects of leg muscle vibration on balance). Eighteen subjects participated in four experimental conditions involving all possible conditions of touch, no-touch, vibration, and no-vibration. Touch stabilized their tandem stance in the absence of vibration and destabilized it in the vibration condition. The other conditions fell in between. 2) We compared the effectiveness of contact cues at the right fingertip in subjects (n=18) assuming a tandem stance under four conditions: 1) fingertip contact, right arm free, and 2) arm immobilized; 3) no fingertip contact and right arm free, and 4) immobilized. In the arm immobilized conditions, the entire arm was rigidly prevented from independent movement by a “Freedom Gunslinger” splint (Alimed, Dedham MA) which coupled the arm to the torso. The results showed that even though immobilizing the arm prevented stable finger contact with a surface the intermittent contact nevertheless stabilized balance as much as continuous contact with the arm not splinted. 3) We measured the time courses of 1) postural stabilization when subjects (n=12) standing in the dark were allowed full sight of their surroundings, and 2) postural destabilization when subjects (n=12) standing in full room illumination were suddenly exposed to complete darkness. The results show that the time course of visual stabilization is about three times longer than that of touch cues at the hand, by contrast postural destabilization occurs more rapidly for loss of visual input than for loss of contact cues at the hand. Together these results emphasize the great value of providing haptic cues at the hand for stabilizing balance and indicate such cues can be intermittent and that they stabilize balance with a short time constant.
Bibliography Type:	Description: (Last Updated: 03/31/2017)
Articles in Peer-reviewed Journals	Bryan AS, Bortolami SB, Ventura J, DiZio P, Lackner JR. "Influence of gravitoinertial force level on the subjective vertical during recumbent yaw axis body tilt." Exp Brain Res. 2007 Nov;183(3):389-97. PMID: 17703287 , Nov-2007
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Books/Book Chapters	Lackner JR, Di Zio P. "Motion sickness." in "Encyclopedia of Neuroscience." Ed. L.R. Squire. Amsterdam ; Boston : Elsevier Academic Press, c2009. p. 989-993. http://dx.doi.org/10.1016/B978-008045046-9.00278-3 , Jan-2009